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Magnetic Moment Measurement of a Coordination Complex

*An experiment in undergraduate
inorganic chemistry*

The measurement of the magnetic moment of a coordination complex containing unpaired electron spins is useful in the establishment of the valency of the metal atom, and in many cases also helps to determine the geometrical structure of the complex (1-3). It is thought to be desirable to introduce students to this important technique early in the sophomore inorganic course at Reed, so that they may, if they wish, make use of magnetic moment measurements in independent laboratory work on the structure and reactivity of complexes during the latter part of this course.

The Gouy method for determining magnetic susceptibility (2, 4, 5) is based on the measurement of the force exerted on a sample by an inhomogeneous magnetic field. The force, dF , exerted in the x direction of an element of matter with volume, dv and mass, dm , by a magnetic field of strength, H , which has a gradient in the x direction, dH/dx , is given by the expression

$$dF = K dv H \frac{dH}{dx} \quad (1)$$

where K is the volume susceptibility. In the Gouy method the sample is contained in a cylindrical tube. The Pyrex tube used in these experiments is simple enough in design so that it can be made by students. The tube has an outside diameter of 8 mm and a length of 15 cm. The tube is placed in a magnetic field¹ which is uniform in a horizontal plane but which decreases along the length of the tube. Tapered magnet pole pieces (available from Alpha Scientific Laboratories) shown in the figure were used to obtain a sizeable uniform field gradient over the length of the tube. For analysis of the data obtained by the Gouy method an integrated form of eqn. (1) is used, found by substituting $dv = A dx$ (where A is the cross-sectional area of the tube) and integrating. The resulting equation is:

$$F = \frac{1}{2}AK(H^2 - H_0^2) \quad (2)$$

where H is the field at the bottom of the Gouy tube, H_0 is the field at the top of the sample in the tube, and F is the force exerted on the sample by the magnetic field.

K of eqn. (2) is the susceptibility associated with the sample and the tube combined, whereas we are interested in the susceptibility of the sample alone. Taking

the force contribution of the tube as a constant,² δ , measured by determining the weight change of the empty tube in a given magnetic field, we have a modified equation.³

$$F = \frac{1}{2}AK'(H^2 - H_0^2) + \delta \quad (2a)$$

where K' is the volume susceptibility of the sample.

We obtain on rearranging eqn. (2a) and making the substitution $F' = F - \delta$:

$$K' = \frac{2F'}{A(H^2 - H_0^2)} \quad (3)$$

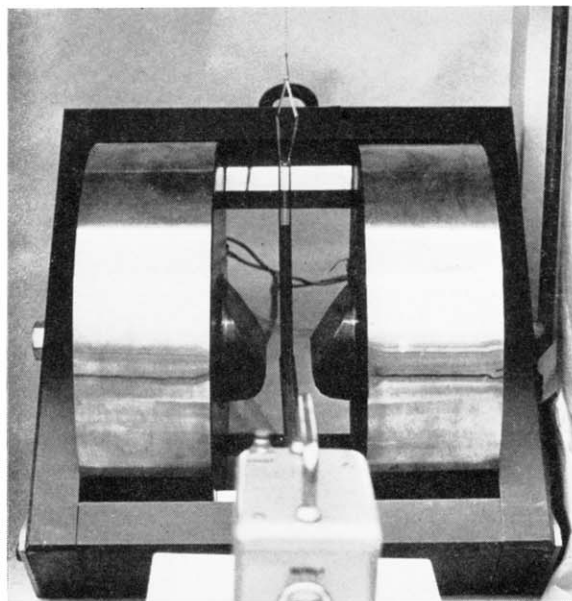
The specific susceptibility, X , is next obtained by the formula

$$X = \frac{K'}{d}$$

where d is the density of the solid sample. It should be mentioned that the density used in this equation is the density of the sample material as it is packed in the tube obtained by dividing the sample mass by the filled volume of the tube. X_M , the gram molecular susceptibility is obtained from the following equation:

$$X_M = MX \quad (4)$$

where M is the mole weight. X_M is corrected for dia-



The Gouy tube and the Gaussmeter probe positioned for measurement.

¹ The magnetic field is generated by an Alpha AL7500 Electro-magnet powered by an AL7500 PS filtered power supply coupled with an AL7500 R current regulator.

² δ is negative since the glass tube is diamagnetic.

³ It should be noted that in work of the utmost accuracy K' must be corrected for the magnetic susceptibility of the air displaced by the sample (the volume susceptibility of air at STP is $+ 0.029 \times 10^{-6}$ cgs units/cc).

magnetic contributions to the susceptibility by making use of tables of empirically determined corrections for atoms, bonds, ions, and ligands. On pages 142 and 143 of Reference (3) there are reasonably complete diamagnetic correction tables. X'_M , the corrected paramagnetic susceptibility, is obtained by adding the total diamagnetic correction C_D (taken as a positive number): $X'_M = X_M + C_D$. The effective magnetic moment, μ_{eff} , is then obtained using the defining expression eqn. (1).

$$\mu_{\text{eff}} = \left(\frac{3k}{N\beta^2} \right)^{1/2} (X'_M T)^{1/2} = 2.828 (X'_M T)^{1/2} \text{ B.M.} \quad (5)$$

where k is Boltzmann's constant, β is the Bohr magneton, N is Avagadro's number, and T is the absolute temperature. B.M. indicates that μ_{eff} is expressed in Bohr magneton units. Equation (5) assumes that the dependence of the susceptibility on temperature obeys the Curie law:

$$X = \frac{C}{T} \quad (6)$$

where C is the Curie constant and T is the absolute temperature. Most magnetically dilute compounds are found to obey the Curie law in the room temperature range, so this is a reasonable assumption.

The simplest theoretical analysis of the results of a magnetic moment measurement, which is reasonably valid for most compounds containing metals of the $3d$ transition series, is to assume that only the unpaired electron spins make a significant contribution to the magnetic moment, the contribution due to the orbital motion of the electron being negligible. In this case the magnetic moment is predicted to have the value given by the following equation:

$$\mu_{\text{eff}} = [n(n + 2)]^{1/2} \text{ B.M.} \quad (7)$$

where n is the number of unpaired electrons. By use of eqn. (7) we can calculate the number of unpaired electrons and obtain the valence of the metal ion.

The Experiment

The purpose of the experiment is to determine the magnetic moment of a transition metal complex by a method of sufficient accuracy to establish the valence of the metal ion. The compound $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$ was chosen for this purpose, since it is easily prepared in high purity (7), the described synthesis resulting in large well-formed single crystals. In addition it has the advantage of having a spin free d^5 configuration which should give a large value for the susceptibility and make possible accurate measurements of the magnetic moment with equipment consisting of an analytical balance of 0.1 mg sensitivity and a magnet which has a peak magnetic field of 3 KG or greater. The d^5 configuration is also favorable for comparison of the value of the experimental magnetic moment with that calculated by eqn. (7), because if we have a d^5 complex where all d -electron spins are unpaired (as in the case of $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$) the total orbital angular momentum will be equal to zero; and, therefore, there should be no orbital contribution to the magnetic moment.

The student is asked to measure the magnetic susceptibility of $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$ by two modifications of the Gouy method.

Since it was thought to be pedagogically desirable to

make direct use of eqn. (3), an experiment was carried out where H and H_0 were measured with a Gaussmeter. In addition it was felt necessary to familiarize the student with a relative method of obtaining X , so the student was asked to prepare a compound of known magnetic susceptibility, $\text{HgCo}(\text{CNS})_4$ (8), and use this compound as a calibrant.

Finally, the student is asked to demonstrate for himself the low intensity associated with a spin forbidden electronic transition by measuring the visible absorption spectrum of $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$. Since this compound has all five $3d$ levels filled with one electron, each with all spins unpaired, transitions of an electron from one d level to another expected in the visible region (6) must go with spin flipping in order that the Pauli exclusion principle be obeyed in the excited state. Since a spin flip is a reasonably slow process compared to the duration of an electronic transition, such transitions are very unlikely and are said to be spin forbidden.

Magnetic Measurements

Direct Method. The figure shows the sample tube (attached by a silver chain to a 0.1 mg sensitivity analytical balance with a hole drilled under one pan and the chain mounted to that pan), the tapered pole pieces of the magnet, and the Alpha AL 100C Gaussmeter probe in position for measurement.

It is important to have the Gouy tube as uniformly packed as possible with the solid sample, since non-uniformity of packing is the major source of error in this method. In the case of $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$ the large crystals were thoroughly ground in a mortar and pestle, the resulting microcrystalline solid was added to the tube in portions of about 0.1 g, and the bottom of the tube was rapped sharply four times after each addition. The same procedure was followed for $\text{HgCo}(\text{CNS})_4$ except that this compound was used in the form in which it was prepared, because the solid consisted of small crystals of rather uniform size.

Equation (3) was used to calculate K' in the direct Gouy method. Forces due to the magnetic field were calculated from weight changes by multiplying them by 980.6 cm/sec^2 , the value of the gravitational constant at Portland, Oregon (9). In a typical experiment where $T = 30^\circ\text{C}$, $H = 4.40 \pm 0.03 \text{ KG}$, H_0 (tube filled to 11.85 cm) = 0.505 KG , and $A = 0.453 \pm 0.003 \text{ cm}^2$ (measured by weighing the tube filled to a given height with water), μ_{eff} was found to be $5.93 \pm 0.07 \text{ B.M.}$ which is in excellent agreement with the value calculated from eqn. (7), $\mu_{\text{eff}} = 5.92 \text{ B.M.}$ The results of almost all student measurements fell in the range $\mu_{\text{eff}} = 5.95 \pm 0.10 \text{ B.M.}$, so that the direct measurement of μ_{eff} gives results which are in good agreement with that expected for a magnetically dilute iron(III) compound obeying Hund's rule.

Relative Method. This method is one of great practical importance because magnetic field measurements can be avoided. Equation (3) is easily modified to a form useful for the relative method. By substituting $K' = dX$ into eqn. (3) and rearranging we have

$$X = \frac{2F'}{d(H^2 - H_0^2)} = \frac{2\Delta m' \times 980.6 \times V}{AW(H^2 - H_0^2)} \quad (8)$$

where d is the sample density, W is the sample weight,

$\Delta m'$ is the weight difference in grams caused by application of the magnetic field to the sample, and V is the volume occupied by the sample. Since the experiment is arranged so that V , H , and H_0 are the same for both the calibrant and unknown measurements, eqn. (8) can be rewritten as

$$X = \frac{\beta' \Delta m'}{W} \quad (9)$$

where β' is a calibration constant which can be evaluated by measuring W and $\Delta m'$ for a compound with a known value of X .

In a typical experiment using $\text{HgCo}(\text{CNS})_4$ as a calibrant ($X = 16.44 (\pm 0.08) \times 10^{-6}$ cgs units at 20°C) $\mu_{\text{eff}} = 5.85 \pm 0.03$ B.M. Since the measurement was made at 30°C , X of the calibrant was corrected to this temperature by use of Curie's Law: $X(30^\circ\text{C}) = X(20^\circ\text{C}) \times \frac{293.2}{303.2}$.

Again we have a value that is in quite close agreement with that predicted by eqn. (7). The fact that the theoretical value does not quite fall within the limit of error predicted for μ_{eff} is most likely explained by a systematic error due to non-uniform packing of the sample. Student results for the relative method were in the range 5.90 ± 0.10 B.M.

It is interesting to note that our values of μ_{eff} for $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$ are in better agreement with theory than the value 5.75 B.M. previously reported in the literature (10).

The visible spectrum of $\text{K}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 3\text{H}_2\text{O}$ in aqueous solution was measured on a Beckman DB spectro-

photometer. The low value of the peak molar absorbance ($\epsilon = 0.9$ at $\lambda = 670 \text{ m}\mu$) compared to values usually observed for transition metal complexes where spin allowed transitions are possible (6) indicates that the transition is spin forbidden.

Acknowledgment

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